

Understanding and Improving Low-Resource Neural Machine Translation with Shallow Features

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Abstract. During neural machine translation (NMT) tasks, we observe that despite the common assumption that increasing encoder depth leads to improved performance, this effect is less pronounced in low-resource scenarios and can even exacerbate overfitting issues. Our comparative analysis between NMT models equipped with shallow and deep encoders reveals that the majority of sentences are more effectively translated by a shallow encoder. Further analysis indicates that these sentences tend to be simpler, suggesting the shallow encoder's ability to capture unique features in simple text. Building on these insights, we introduce **NATASHA**, a novel training strategy that enhances the capabilities of deep models in low-resource **N**eural m**A**chine **T**r**A**nslation with **SH**allow fe**A**tures extracted through sequence-level knowledge distillation from the shallow model. Experimental results on five low-resource NMT tasks show that NATASHA consistently improves over strong baselines by at least 1 BLEU point. Furthermore, when combined with other regularization methods, NATASHA achieves leading-edge performance on the IWSLT14 De-En translation task. Further analysis of our method's effectiveness reveals that integrating shallow features reduces the complexity of the training data, facilitating deep models in learning patterns and features within simple text during the early stages of training. This unleashes the deep model's ability to learn representations of low-frequency words and long sentences, thereby enhancing overall performance.

Keywords: Low-Resource Neural Machine Translation · Encoder Depth · Shallow Features

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1 Introduction

Low-resource neural machine translation (NMT) faces challenges due to the scarcity of parallel data [30]. Previous studies have explored various approaches such as data augmentation [6,38], unsupervised learning [12], and pre-training models [3,25] to address these challenges. While increasing encoder depth has been effective in enhancing NMT performance [13,18,34], its full potential remains largely unexplored in low-resource NMT scenarios. Studies [4,30] indicate that low-resource NMT is sensitive to model depth and other hyper-parameters. However, there is a limitation in understanding the impact of model depth on NMT performance in low-resource settings. Given that Kasai et al. [8] demonstrate the significant benefits of deepening encoders over decoders, our focus is primarily on encoder depth.

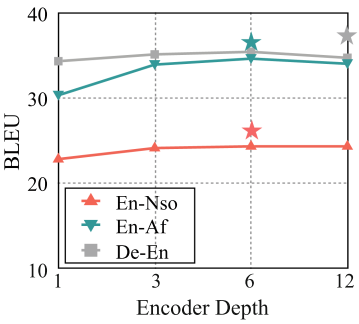


Fig. 1. The BLEU scores of the three low-resource NMT tasks vary with different encoder depths. ⋆ denotes the BLEU scores achieved by our NATASHA method.

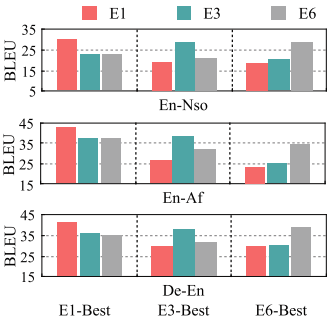


Fig. 2. Significance analysis of translation performance (BLEU scores) using multiple models with varying encoder depths on the same group of test data.

To address gaps in current research, in this paper, we systematically investigate the effect of encoder depth on low-resource NMT models. We evaluate NMT models with varying encoder depths (i.e., 1, 3, 6, and 12 layers) on three representative low-resource NMT benchmarks. As shown in Fig. 1, while increasing depth to 6 layers improves performance over 1 and 3 layers, further improvements to 12 layers are minimal. Over-abstraction of source representation, as suggested by Liu et al. [19], can lead to model hallucinations. This motivates us to compare the performance of source sentences in detail by investigating multiple low-resource NMT models with varying encoder depths.

Surprisingly, we observe that shallow or medium-depth encoders outperform deep ones for almost half of the source sentences. Significance analysis reveals a substantial performance gap between shallow and deep encoders on the same group of data. Further linguistic analysis of the source sentence indicates that they are typically short and have small embedding norms, **suggesting the shallow encoder's ability to capture distinctive features in simple text.** This

insight sheds light on the potential of leveraging shallow features embedded in shallow encoders to improve the shortcomings of deep models in handling simple text, and further unleash the potential of deep models in handling complex text.

Building on the insights above, we propose a simple yet effective training strategy **NATASHA** to improve low-resource **N**eural m**A**chine **T**r**A**nslation with **SH**allow fe**A**tures. The core idea is to enhance the capabilities of deep models by integrating shallow features derived from the shallow model. We employ sequence-level knowledge distillation proposed by Kim et al. [9] to extract these features. Our contributions can be summarized as follows:

1) We systematically investigate the effect of encoder depth on low-resource NMT models to address gaps in existing research. Our findings reveal that models with a shallow encoder better handle simple text.
2) We propose NATASHA, a simple and effective method for training a deep NMT model using the shallow features extracted from a shallow NMT model.
3) NATASHA consistently improves performance across five low-resource NMT tasks by at least 1 BLEU point, achieving a leading-edge BLEU score of 38.2 on the IWSLT14 De-En task when combined with the representative data augmentation method CipherDA [7].
4) Further analysis of the reasons for NATASHA's effectiveness reveals that incorporating shallow features simplifies training data, aiding deep models in learning patterns and features within simple text during the early stages of training. This unleashes the potential of deep models in representing complex text such as low-frequency words and long sentences, thereby enhancing overall performance.

2 Related Work

2.1 Low-Resource NMT

Neural networks with a large number of parameters also require a massive amount of training data for the NMT task. However, low-resource NMT tasks lack abundant parallel data [30]. To improve translation performance in low-resource scenarios, previous studies mainly explore from three perspectives: 1) expanding parallel data [28,38], 2) exploring various unsupervised ways to learn from unlabeled monolingual data [12], and 3) sharing knowledge between high- and low-resource languages by multi-task learning [5,27,37] or initializing NMT models with parameters from pre-training models [3,14,17,25].

While increasing encoder depth has been effective in enhancing NMT performance [13,18,34], the parameter configuration of low-resource NMT has received limited attention in previous research. Studies by Biljon et al. [4] and Sennrich et al. [30] have highlighted the sensitivity of low-resource NMT to hyperparameters. Specifically, Biljon et al. [4] discuss the optimal transformer depth for low-resource translation, emphasizing the critical role of model depth. Building upon these findings, our study focuses on investigating the impact of encoder depth in low-resource scenarios, providing valuable insights that complement existing research.

2.2 Layer Depth of NMT

Model depth has been proven to be a significant factor in NMT research [21, 22]. With the impressive performance demonstrated by the Transformer model [33], many studies have made significant efforts to train deeper Transformer models [13,34]. It has been shown that increasing the encoder depth brings more noticeable performance improvements than increasing the decoder depth for the Transformer architecture [8], leading subsequent works to primarily focus on deepening the encoder [11,35].

Despite these findings, it remains unexplored whether low-resource NMT tasks can benefit more from deepening the encoder depth, which is the main focus of this paper.

3 Understanding Encoder Depth of Low-Resource NMT

3.1 Preliminary

NMT aims to translate text from a source language $\mathbf{x}$ to a target language $\mathbf{y}$, to maximize the log probability of $\mathbf{y}$ given $\mathbf{x}$, i.e., $\text{argmax} \log P(\mathbf{y}|\mathbf{x})$. To bridge the representation gap between source and target languages, the widely used approach in NMT is the encoder-decoder architecture [32,33]. Among them, Transformer [33] is a mainstream architecture powered by the self-attention mechanism. It consists of N stacked encoder layers $\mathcal{E}_N = \{\mathcal{E}_N^{(1)}, ..., \mathcal{E}_N^{(N)}\}$ and M stacked decoder layers $\mathcal{D}_M = \{\mathcal{D}_M^{(1)}, ..., \mathcal{D}_M^{(M)}\}$. The model forward process can be expressed as follows:

$$\mathbf{h}(\mathbf{x}) = \mathcal{E}_N(\text{Embed}(\mathbf{x})) \tag{1}$$

$$\hat{\mathbf{y}} = \text{Proj}(\mathcal{D}_M(\widetilde{\mathbf{y}}, \mathbf{h}(\mathbf{x}))) \tag{2}$$

Here, $\text{Embed}(\cdot)$ represents the combination of word embedding and positional embedding, $\widetilde{\mathbf{y}}$ denotes the shifted right $\mathbf{y}$, and $\text{Proj}(\cdot)$ denotes the projection of decoder representation. The model parameters are optimized using the gold truth $\mathbf{y}$ and the model output $\hat{\mathbf{y}}$.

3.2 Experimental Setup

Datasets. We conduct experiments on the three language pairs to evaluate the performance: Autshumato English-Northern Sotho Parallel Corpora[1] (En-Nso; 30K instances), Autshumato English-Afrikaans (En-Af; 50K instances), and IWSLT14 German-English (De-En; 160K instances). These language pairs represent small-, medium-, and large-scale corpora for low-resource translation, respectively. For the En-Nso and En-Af, we follow the setups from [1] and use an 8K vocabulary based on a joint dictionary with byte pair encoding (BPE; [29]). For the De-En, we use IWSLT14 data splits and follow the setups from [15].

[1] https://github.com/LauraMartinus/ukuxhumana.

Table 1. The ratio of translations where the shallow model outperforms the deep one. "ex." is the abbreviation of "exceed" while "$\mathcal{E}_{\text{shallow}}$ ex. $\mathcal{E}_{\text{deep}}$" denotes that a shallow model outperforms the deep one. To mitigate the impact of random factors, we train two additional models by initializing the parameters with different random seeds and then averaging the results.

$\mathcal{E}_{\text{shallow}}$ ex. $\mathcal{E}_{\text{deep}}$	En-Nso			En-Af			De-En		
	Train	Dev	Test	Train	Dev	Test	Train	Dev	Test
$\mathcal{E}_1$ ex. $\mathcal{E}_3$	51%	50%	54%	62%	55%	56%	58%	61%	62%
$\mathcal{E}_3$ ex. $\mathcal{E}_6$	62%	49%	56%	91%	55%	63%	59%	62%	63%
$\mathcal{E}_1$ ex. $\mathcal{E}_6$	42%	48%	54%	62%	52%	52%	51%	59%	60%

Models and Hyperparameters. We implement the Transformer model [33] using the *fairseq-py*[2] toolkit [23]. For all tasks, we use the Transformer with the following configurations: 512 hidden dimensions, 1,024 FFN (Feed-Forward Network) dimensions, and 4 attention heads for both the encoder and decoder. The experiments are conducted with a 6-layer decoder and variable encoder depth. To update the model parameters, we use the Adam optimizer [10] with the following hyperparameters: $\beta_1 = 0.9$, $\beta_2 = 0.98$, $\epsilon = 10^{-9}$. The peak learning rate is set to 0.0005 and we apply the inverted squared decay [33] with 4K warm-up steps. We train all models using four NVIDIA V100 GPUs, and each batch contains 4096 tokens. During model validation, we evaluate the model on the validation set at each epoch using the BLEU score. If the BLEU score does not increase after 10 consecutive validation times, we stop the training process. During the inference stage, we use a beam size of 5 for all tasks.

Evaluation Metrics. To evaluate the performance of the model, we use the widely-used BLEU [24] metric. BLEU is an n-gram based metric that measures the precision rate to assess the coverage of reference n-grams in the model's hypothesis. We report case-sensitive tokenized BLEU score using *multi-bleu.perl*[3].

3.3 Observing Performance Drop in Deep Encoder Depth

Generally, increasing the depth of the encoder improves translation performance for high-resource language pairs but may lead to overfitting in low-resource datasets. In this study, we conduct experiments in low-resource scenarios and observe the performance. Figure 1 displays the BLEU scores of three NMT tasks on the test set with different encoder depth settings.

The NMT models are denoted as $\mathcal{E}_i$, representing a model with an i-layer encoder. As shown in Fig. 1, the BLEU score of $\mathcal{E}_{12}$ model is similar to that of $\mathcal{E}_3$

[2] https://github.com/pytorch/fairseq/.

[3] https://github.com/moses-smt/mosesdecoder/blob/master/scripts/generic/multi-bleu.perl.

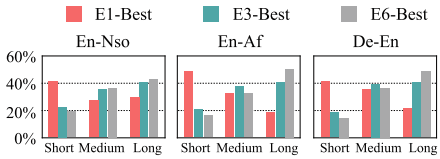
and $\mathcal{E}_6$ models. This suggests that a shallow encoder is sufficient for translation tasks with limited parallel data. Liu et al. [19] observe that excessive abstraction of source sentences leads to hallucinations in NMT models. **This inspires us to compare the performance of source sentences in detail by investigating multiple low-resource NMT models with varying encoder depths.**

For the Majority of Source Sentences a Shallow Encoder Outperforms a Deep One. To assess the performance of source sentences across multiple models with varying encoder depths, we conduct experiments using different encoder configurations and thoroughly analyze the results. Specifically, we calculate the percentage of sentences where the shallow encoder model outperforms the deep encoder model in terms of translation quality, as measured by sentence-level BLEU scores.

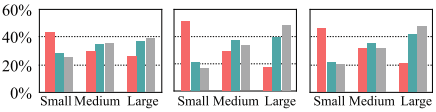
As shown in Table 1, the results from the three translation tasks indicate that approximately half of the sentences are translated more effectively by a shallow encoder compared to a deep one. In the most extreme case, 91% of the training data demonstrates improved performance with a shallow encoder.

There Is a Significant Performance Gap Between Shallow and Deep Encoders When Applied to the Same Group of Data. Although many sentences do not require a deeper encoder, there is concern about whether a shallow encoder is significantly better than a deep one. It might not be surprising if a shallow encoder performs slightly better than a deep one (e.g., by more than 0.05 BLEU points). To address these doubts, we conduct a significance analysis of translation performance on multiple models with varying encoder depths on the same group of test data. Specifically, we categorize the test data into three groups based on optimal translation performance and encoder depth (e.g., $\mathcal{E}_1$-Best denotes the group of data achieving the best performance using the $\mathcal{E}_1$ model). Figure 2 shows the BLEU score margins for each group, with the gap between the lowest and highest performance reaching up to 10 points. This substantial margin dispels our doubts.

A Shallow Encoder Performs Better in Handling Simple Text. To better understand the reasons behind the performance degradation of deep encoders, we conduct a linguistic analysis of the above three groups' source sentences, considering linguistic features like sentence length and word embedding norms. We compare the sentence length and word embedding norm distribution across the three groups' source sentences. As shown in Fig. 3(a), $\mathcal{E}_1$-Best group includes most short sentences while $\mathcal{E}_6$-Best group contains most long sentences. This demonstrates that longer sentences have more complex structures, benefiting from layer-to-layer abstraction. From Fig. 3(b), we observe that sentences with smaller norms are better suited for representation by shallow encoders.



(a) Sentence Length



(b) Norm of Word Embeddings

Fig. 3. Linguistic analysis of the performance drop in deep encoders involves comparing source sentences across three groups by length and word embedding norms. The shortest 1/3 of sentences are categorized as the "Short" group, while the "Medium" and "Long" groups follow similar grouping methods. Similarly, sentences with the lowest embedding norms are divided into "Small" and "Medium" groups, with the top 1/3 to 2/3 belonging to the latter. The remaining sentences are classified into the "Large" group.

4 Our Method

4.1 NATASHA

Motivation. Based on the detailed analysis in Sect. 3, we find that a shallow encoder excels in handling simple text by capturing its unique features effectively. Building upon this insight, we introduce NATASHA, a training strategy that trains a deep model using shallow features extracted from a shallow model to acquire essential features for processing simple text, thereby avoiding overfitting.

Implementation. As depicted in Fig. 4, we initially distill the original training data with a shallow NMT model and subsequently blend this with the original training data. To elaborate, let's consider a parallel corpus $\mathcal{D} = \{< \mathbf{x}^n, \mathbf{y}^n >\}_{n=1}^{N}$, where $\mathbf{x}$ and $\mathbf{y}$ represent source and target sentences, respectively. Firstly, we train a shallow model $\mathcal{M}_{shallow}$ on the training corpus $\mathcal{D}$ and utilize it to translate the source sentences $\{\mathbf{x}^n\}_{n=1}^{N}$ to the target sentences $\{\mathbf{y}_{shallow}^n\}_{n=1}^{N}$. Through the sequence-level knowledge distillation technique [9], the unique features are effectively embedded into the new corpus $\mathcal{D}_{shallow} = \{(\mathbf{x}, \mathbf{y}_{shallow})\}_{n=1}^{N}$. In the subsequent phase, we merge $\mathcal{D}$ with $\mathcal{D}_{shallow}$ to construct a fresh parallel corpus for the deep NMT model training. This process enables the deep model to assimilate not only the features from the original training data but also the unique shallow features from the distillation data, thus facilitating the breakthrough of its performance bottleneck.

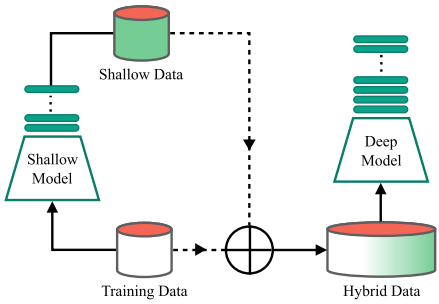


Fig. 4. The overall framework of NATASHA.

4.2 Evaluation

Setup. For the model settings and implementation, we replicate previous experiments using the same Transformer settings for En-Nso, En-Af, and De-En. Additionally, we extend our evaluation to include two additional low-resource NMT tasks: Turkish-English (Tr-En; 190K instances) and WMT16 Romanian-English (Ro-En; 610K instances). For these tasks, we use Transformer-base settings with 512 hidden dimensions, 2,048 FFN dimensions, and 8 attention heads for both the encoder and decoder. For Tr-En, we use the WMT18 parallel data from the SETIMES22[4] corpus and employ the official WMT17 and WMT18 test sets as the development and test sets, respectively. For Ro-En, we follow the setups of [19] and use the WMT16 data splits, with the newsdev16 set as the development set and newstest16 as the test set.

Regarding evaluation metrics, we report the 4-gram BLEU score using the *multi-bleu.perl*[5] script, consistent with previous studies [1,15,19]. Additionally, to ensure comprehensiveness, we report Translation Edit Rate (TER) scores[6] [31], which measure the edit distance between system translations and references.

Main Results. In Table 2, we report the results of NATASHA on all five translation tasks compared to the baseline. We implement the vanilla NMT models with variable encoder depth and conduct experiments on $\mathcal{E}_6$ and $\mathcal{E}_{12}$ models for the proposed method. From Table 2, it is evident that models trained with NATASHA exhibit improvements of 0.3–2.3 BLEU points over their baselines and approximately 1.0 BLEU point improvement over strong baselines across all tasks under the same experimental conditions. This indicates the universal effectiveness of our approach for low-resource datasets of all sizes.

Furthermore, we observe that the optimal encoder depth for the low-resource En-Nso and En-Af translation tasks is 6 layers, while for others it is 12 layers, suggesting that excessively deep encoders are unnecessary for extremely

[4] https://opus.nlpl.eu/SETIMES2.php.
[5] https://github.com/moses-smt/mosesdecoder/blob/master/scripts/generic/multi-bleu.perl.
[6] https://github.com/mjpost/sacrebleu.

Table 2. Translation performance on different translation benchmarks. NATASHA-$\mathcal{E}_6$ means training an NMT model of 6 encoder layers with the shallow features extracted from $\mathcal{E}_3$, while the shallow features for training NATASHA-$\mathcal{E}_{12}$ of 12 encoder layers are extracted from $\mathcal{E}_6$.

Model	30K En-Nso		50K En-Af		160K De-En		190K Tr-En		610K Ro-En	
	BLEU	TER	BLEU	TER	BLEU	TER	BLEU	TER	BLEU	TER
Vanilla NMT models										
$\mathcal{E}_1$	22.8	69.2	30.3	59.1	34.3	52.2	16.3	74.4	33.0	55.3
$\mathcal{E}_3$	24.1	68.0	33.9	54.9	35.1	51.6	17.6	72.6	34.0	54.6
$\mathcal{E}_6$	24.3	68.8	34.6	54.4	35.4	51.1	18.1	72.1	34.1	54.8
$\mathcal{E}_{12}$	24.3	67.7	34.0	55.2	34.7	51.5	18.0	71.8	34.3	54.2
Our Approach										
NATASHA-$\mathcal{E}_6$	**25.9**	**65.0**	**35.9**	52.7	36.4	49.9	18.7	70.3	34.4	53.9
NATASHA-$\mathcal{E}_{12}$	25.4	65.7	35.5	**52.4**	**37.0**	**49.5**	**19.4**	**70.0**	**35.3**	**53.5**

Table 3. Comparison to Existing Works on the IWSLT14 De-En task.

Model	BLEU
Vanilla Transformer [33]	35.4
BERT-Fused [40]	36.1
R-Drop [15]	37.3
CipherDAug [7]	37.5
NATASHA-$\mathcal{E}_{12}$	37.0
NATASHA-$\mathcal{E}_{12}$ w/ R-Drop	37.7
NATASHA-$\mathcal{E}_{12}$ w/ CipherDAug	**38.2**

Table 4. Ablation Study. The BLEU scores of model variants on the IWSLT De-En task.

Model	BLEU
$\mathcal{E}_6$	35.4
$\mathcal{E}_{12}$	34.7
$\mathcal{E}_{12}$ w/ Data Upsampling	35.0
$\mathcal{E}_{12}$ w/ Self Distillation	35.2
$\mathcal{E}_{12}$ w/ Tagging Shallow Data	36.5
NATASHA-$\mathcal{E}_{12}$	**37.0**

low-resource NMT tasks. However, all NATASHA-$\mathcal{E}_{12}$ models outperform $\mathcal{E}_6$ models when increasing the encoder depth, indicating the potential of training deeper models in low-resource scenarios using NATASHA. Additionally, TER scores exhibit a similar trend to BLEU scores, demonstrating the generality of NATASHA.

Comparison to Existing Works. To further validate NATASHA's effectiveness, we compare its translation performance with existing approaches. As shown in Table 3, NATASHA achieves results comparable to other well-established methods that use pretraining or regularization techniques. When combined with the representative regularization method R-Drop [15] and data augmentation method CipherDAug [7], NATASHA further improves by 0.7 and 1.2 BLEU points, respectively, demonstrating its complementarity to these approaches.

4.3 Ablation Study

We conduct an ablation study to assess the impact of different components of our method on the final results.

Effect of Data Upsampling. Since our method doubles the training data, we directly performed data upsampling on the training set to ensure that the improvement in our experimental results is not solely due to data expansion. The results in Table 4 indicate only a slight improvement when using the doubled training set directly, compared to the baseline. This suggests that while increasing the training set size may lead to slight performance improvements in low-resource NMT tasks, it is not sufficient for significant progress. We attribute this to the homogenization of the training data, lacking additional distinguishing features from the original data. Notably, NATASHA outperformed the baseline model by 2.3 BLEU points, indicating its ability to learn more features from hybrid data and highlighting the effectiveness of shallow features.

Effect of Self Distillation. Given that our method utilizes distillation data from a shallow model, we replace the shallow model with a deep model itself to ensure that the effectiveness of our method does not solely come from data distillation. Specifically, we initially train a $\mathcal{E}_{12}$ model and use it to translate the training set and then retrain a new $\mathcal{E}12$ model by combining the distillation data with the original data. The results in Table 4 show a marginal improvement over the baseline when replacing the shallow model with the deep model itself, confirming the effectiveness of shallow features.

Effect of Tagging Shallow Data. Shallow features constitute a critical element of our method. To investigate the impact of weakening these features on performance, we use tagging to differentiate distillation data from the original training data. This method, commonly utilized to distinguish translationese from the original target text, serves to prevent overfitting to translationese [2,20,26]. In our context, we leverage this technique to mitigate overfitting to shallow features. As evidenced in Table 4, tagging the shallow data leads to a degradation in performance, thereby **highlighting the positive effect of shallow features in training deep models in low-resource scenario.**

4.4 Analysis

In this section, we aim to delve deeper into how shallow features positively influence the training of deep models and analyze the reasons for our approach's effectiveness. We focus on the IWSLT De-En task, as it serves as a representative example of low-resource NMT tasks.

In deep learning, gradient norms are useful for understanding and improving the model training process the model training process [16,36]. Therefore, we compare NATASHA with vanilla $\mathcal{E}_6$ based on gradient norms during training. Specifically, we segment the training phase into two stages, focusing on the

Table 5. The translation accuracy of words categorized by frequency and the BLEU scores of sentences categorized by length. Word accuracy is categorized by frequency into "Low" (less than 50 times), "Medium" (50–200 times), and "High" (over 200 times) groups. Sentence BLEU scores are categorized by length into "Long" (over 50 words), "Medium" (30–50 words), and "Short" (under 30 words) groups.

Stage	Model	BLEU		Word Frequency			Sentence Length		
		Valid	Test	Low	Medium	High	Long	Medium	Short
Early	$\mathcal{E}_6$	30.9	30.2	34.7	53.3	67.7	26.0	28.0	32.0
	NATASHA-$\mathcal{E}_{12}$	31.7	31.1	$32.9_{-1.8}$	$53.5_{+0.2}$	$68.5_{+0.8}$	$24.5_{-1.5}$	$28.6_{+0.6}$	$33.4_{+1.4}$
Final	$\mathcal{E}_6$	36.0	35.4	45.3	60.0	70.6	31.9	33.4	37.0
	NATASHA-$\mathcal{E}_{12}$	37.3	37.0	$47.4_{+2.1}$	$61.0_{+1.0}$	$71.4_{+0.8}$	$33.1_{+1.2}$	$34.6_{+1.2}$	$38.4_{+1.4}$

intersection point of the global gradient norms, to identify the advantages and improvements that NATASHA provides over the baseline model at each stage.

As shown in Table 5, NATASHA outperforms the vanilla $\mathcal{E}_6$ at the early stage, the disparity in performance is primarily observed in high-frequency words and short sentences. It can be interpreted that the use of distillation data reduces the complexity of the training data, as reported in Zhou et al. [39], particularly for high-frequency words and short sentences, making it easier to learn patterns and features within the data, thus improving performance. Furthermore, we observe that the performance improvement at the later stage mainly comes from low-frequency words and long sentences. This indicates that **the effective learning of shallow features aids the deep model in representing complex features and unlocking their translation potential for low-frequency words and long sentences**.

5 Conclusion

This study thoroughly compares how different encoder depths affect the performance of source sentences in low-resource NMT. Surprisingly, our findings reveal that the majority of source sentences in low-resource NMT are translated more accurately by a shallow encoder than a deep one, especially those exhibiting characteristics of simple text. Building upon this insight, we introduce NATASHA, a straightforward yet effective method for leveraging the superior performance of shallow encoders in handling simple text. Notably, NATASHA requires no model adjustments or additional corpora, seamlessly integrating with other robust low-resource NMT methods. Across five tasks, it consistently outperforms strong baselines by at least 1 BLEU point, achieving leading-edge results on the IWSLT14 De-En translation task. Further analysis reveals that this improvement primarily stems from enhanced translation accuracy of low-frequency words and long sentences, indicating NATASHA's potential to overcome the fundamental bottleneck of low-resource NMT.

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